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Date: 18/03/2026
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Agentic AI Fundamentals Quiz

Section A: Multiple Choice Questions (MCQs)

1. What is Agentic AI?

- A. AI that only performs predefined tasks
 - B. AI systems that act autonomously to achieve goals
 - C. AI used only for data storage
 - D. AI that requires constant human supervision
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2. Which of the following best describes an “agent” in AI?

- A. A database
 - B. A program that perceives and acts upon an environment
 - C. A neural network only
 - D. A hardware device
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3. What is the primary goal of an AI agent?

- A. To maximize randomness
 - B. To achieve a defined objective or goal
 - C. To reduce memory usage
 - D. To store large datasets
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4. Which component is responsible for deciding actions in an agent?

- A. Sensor
 - B. Actuator
 - C. Policy/Decision-making module
 - D. Memory storage
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5. In Agentic AI, “planning” refers to:

- A. Collecting data
- B. Creating a sequence of actions to achieve a goal
- C. Training a model
- D. Visualizing results

6. Which of the following is NOT a typical agent role?

- A. Perception
- B. Reasoning
- C. Execution
- D. Compilation**

7. Reasoning in AI agents helps to:

- A. Store information
- B. Make logical decisions based on data**
- C. Increase hardware speed
- D. Reduce dataset size

8. What does “execution” mean in Agentic AI?

- A. Designing algorithms
- B. Running planned actions in the environment**
- C. Training a model
- D. Debugging code

9. Which type of agent uses condition-action rules?

- A. Learning agent
- B. Model-based agent
- C. Simple reflex agent**
- D. Utility-based agent

10. Multi-agent systems involve:

- A. Only one AI system
 - B. Multiple agents interacting with each other**
 - C. Only human interaction
 - D. Only hardware devices
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Section B: Short Answer Questions

11. Define Agentic AI in your own words.

Agentic AI refers to AI systems that can autonomously perceive, reason, plan, and act to achieve specific goals without continuous human intervention.

12. List any three roles of an AI agent.

1. Perception 2. Reasoning 3. Execution

13. What is the difference between planning and execution?

Planning: Deciding the sequence of actions needed to achieve a goal

Execution: Actually performing those actions in the environment

14. Explain the importance of reasoning in AI agents.

Reasoning allows an AI agent to analyze data, make logical decisions, and adapt to new situations, improving accuracy and effectiveness.

15. What are sensors and actuators in an agent system?

Sensors: Collect data from the environment (e.g., camera, microphone)

Actuators: Perform actions (e.g., move robot, display output)

Section C: Application-Based Questions

16. A delivery robot must pick up packages and deliver them efficiently.

- Identify the **agent roles** involved.

Perception, Planning, Reasoning, Execution

17. An AI chatbot analyzes user queries and responds intelligently.

- Explain how **reasoning** is used. **The chatbot uses reasoning to understand user intent, analyze context, and generate meaningful responses, instead of giving fixed replies.**

18. Design a simple agent for a **smart home system**.

- What would be its inputs, outputs, and goals?

Inputs: Temperature, motion sensors, user commands

Outputs: Turn lights on/off, adjust AC, security alerts

Goals: Comfort, energy efficiency, safety

19.

Explain how **planning + reasoning + execution** work together in a self-driving car.

Planning: Decide route and maneuvers

Reasoning: Analyze traffic, obstacles, rules

Execution: Control steering, braking, acceleration

20.

Why is autonomy important in Agentic AI? Give a real-world example.

Autonomy enables AI to **operate independently and make real-time decisions**.

Example: Self-driving cars adjusting speed without human input



Section D: Higher-Order Thinking

21. Compare Agentic AI with traditional rule-based systems.

Agentic AI: Adaptive, learns, goal-driven, autonomous

Rule-based: Fixed rules, no learning, limited flexibility

22. What challenges might arise in multi-agent systems?

Communication issues

Coordination complexity

Conflicts between agents

Scalability problems

23. How can Agentic AI improve real-world decision-making?

Agentic AI can analyze large data, predict outcomes, and make **faster, more accurate decisions** in areas like healthcare, finance, and logistics.

24. Explain how feedback helps improve agent performance.

Feedback helps agents **learn from past actions**, improve decisions, and adapt to changing environments (similar to reinforcement learning).

25. Can Agentic AI function without human input? Justify your answer.

Yes, it can operate autonomously, but **initial design, training, and monitoring by humans are still required** for safety and accuracy.